AI DESK COMPANION

Preliminary CAD Model & Engineering Review



Revision: Concept CAD R0.1 **Units:** mm **Purpose:** packaging and design review, not manufacturing release

Important limitation: This package is a preliminary geometric model built from stated assumptions and component envelopes. It is not a verified production CAD design. Exact component models, material properties, fasteners, battery specification, servo models, tolerances and physical test results are not frozen.

1. Executive summary

The model represents a compact, modular desk companion with a detachable AI head, torso, two structural legs and broad feet, articulated arms with grippers, and a head-to-body dock. The head is intended to operate independently; the body adds expressive movement when connected.

The generated geometry contains 75 listed components and reports 0 invalid shapes in the geometry check. The nominal assembly bounding box is 178.0 × 100.2 × 264.1 mm (X × Y × Z). The nominal height is therefore 264.1 mm, below the initial 270 mm target.

A valid-shape count is only a geometric sanity check. It does not prove the model is watertight, printable, mechanically strong, interference-free as an assembly, or safe to operate.

2. Model files and intended use

File	Use / limitation
AI_Desk_Companion_Assembly.step	Full assembly exchange model; inspect in FreeCAD, Fusion 360, SolidWorks or another STEP-capable CAD tool.
AI_Desk_Companion_Exploded.step	Exploded presentation assembly for understanding module separation.
Module_Head.step	Head module subassembly.
Module_Body.step	Torso/body subassembly.
Module_Legs.step	Structural legs.
Module_Feet.step	Left/right foot housings.
Module_Arms.step	Articulated arm/gripper assembly.
AI_Desk_Companion_Assembly.stl	Whole-assembly visual/mesh export. Do not slice as one print; components overlap and need separate part exports.
assembly_view.scad	OpenSCAD assembly-view source included for review; not a replacement for editable feature history in a parametric CAD package.
geometry_manifest.csv	Component names and nominal geometry metadata.
geometry_check.json	Reported bounding box and geometry checks.
cad_preview_iso.png / orthographic_views.png	Preview images for visual review.

3. Nominal geometry

Parameter	Current nominal value	Status
Overall assembly envelope	178.0 × 100.2 × 264.1 mm	Calculated from model bounds
Head outer envelope	104 × 77 × 80 mm	Preliminary shell target
Torso envelope	70 × 52 × 51 mm	Preliminary packaging
Leg spacing (centre-to-centre)	62 mm	Assumption
Structural leg envelope	18 × 22 × 100 mm	Assumption; stress not checked
Each foot envelope	58 × 92 × 16 mm	Assumption; support margin needs testing
Head shell wall thickness	2.5 mm	Assumption; material/print orientation not specified

The drawing's original overall-width and depth labels were ambiguous. The CAD bounds are more useful for packaging, but do not define a finished product envelope once cable protrusions, arm motion, connector access and manufacturing clearances are included.

4. Component packaging assumptions

The current assembly uses bounding-envelope placeholders for internal electronics. The exact Pi 5 PCB, cooler, camera module, display, microphone board, speaker, battery and body controller were not supplied as manufacturer CAD files. Their locations are conceptual, and the package must not be treated as proof that the hardware fits.

Head module intended contents: Raspberry Pi 5, active cooling, display/face window, camera, microphone array, speaker, protected battery pack, physical camera privacy switch and docking contacts. Body module intended contents: ESP32-class controller, actuator drivers, arm servos, wiring, dock detection and a separately powered actuator rail.

Use vendor STEP files and measured part dimensions in the next CAD revision. Check camera field of view, lens-to-window clearance, display mounting, battery removal, cooling airflow, port access, cable bend radii, fastener access and assembly order.

5. Physics and structural review

Stability: the centre of gravity must remain inside the support polygon formed by the feet in every allowed arm/head pose. The current model does not contain a verified mass distribution, so no certified tipping margin can be stated.

Tipping check: compare restoring moment $m \cdot g \cdot d$ against disturbance moment $F \cdot h$. Here d is the horizontal distance from the projected centre of gravity to the tipping edge and h is the height where a horizontal force acts. Measure the assembled centre of gravity and test asymmetric arm poses before finalizing the feet.

Legs and feet: reinforce the torso-to-leg and leg-to-foot interfaces. Use a defined load path rather than relying on cosmetic shell walls. Material, print orientation, infill, inserts, fasteners and expected handling loads are not specified, so strength and fatigue remain unverified.

Arms: determine joint torque from the arm-link masses, maximum reach, intended payload, speed and acceleration. For a payload alone, static torque is $\tau = m \cdot g \cdot r$; arm self-weight, friction and dynamic loads must also be included. Do not select servos by stall torque alone.

Scope: the model's legs are structural supports, not walking legs. It should be treated as a stationary desk robot with expressive head/arm motion unless a different locomotion system is designed.

6. Electrical, power and thermal review

The Raspberry Pi 5 recommended power configuration is 5 V / 5 A. Confirm the chosen power supply, battery topology and regulator can sustain the real load, including transient demand. Do not feed servo or motor current from the Pi's logic rail.

Measure current and power during idle, listening, speaking, vision processing, arm movement and startup. Runtime cannot be reliably calculated from mAh alone: use nominal pack voltage, usable energy, conversion efficiency and measured average load.

Use separate protected head and body power domains for the initial modular prototype. Define charging, over-current/over-temperature protection, grounding and motor-noise mitigation. Do not tie separate battery rails together through the dock without a purpose-designed power-sharing circuit.

Thermal: provide a defined inlet/outlet path for the Pi cooler and keep the battery away from hot components. Verify sustained workload temperatures inside the closed enclosure. Microphone placement must account for fan noise and speaker echo.

Docking: magnets should align/retain the head but should not be the only load-bearing structure. Add keyed geometry that carries shear and prevents incorrect orientation. Define pogo-pin pinout, contact current, ESD protection, ground reference and disconnect behaviour. The Pi 5 USB-C port is power input; data should use an appropriate USB host connection or a separately designed serial interface.

7. Preliminary mass budget

Module	Current concept target	Confidence
Head module	250–350 g	Estimate only
Body/torso module	300–450 g	Estimate only
Legs and feet (pair)	200–350 g	Estimate only
Complete robot	750–1,150 g	Unverified sum

The mass targets are not based on a complete bill of materials. Add measured weights for every board, battery, servo, fastener, cable, connector, shell and insert; then compute centre of gravity from the component masses and positions.

8. Required verification before fabrication

- Import the STEP into the target CAD package and confirm units, orientation and component count.
- Replace electronics placeholders with manufacturer STEP models; perform assembly interference checks.
- Define tolerances, wall thicknesses, inserts, fasteners, material and manufacturing process.
- Calculate arm joint torque and specify actuator models from actual load cases.
- Compute centre of gravity and tipping margin for neutral and worst-case arm/head poses.
- Check structural legs, foot joints, dock retention and gripper loads.
- Finalize battery chemistry/topology, charging/protection, regulators, current budget and thermal path.
- Define docking electrical pinout and test disconnect/fault response.
- Perform physical fit, runtime, temperature, stability and repeated docking tests.

9. Known limitations

The STEP and STL files are generated concept exports, not a manufacturing release. The combined STL represents overlapping assembly components and must not be sent to a slicer as one print. Use the separate component files and verify them before printing; the supplied individual STL set includes illustrative envelopes and may not represent production-ready part geometry.

No finite-element stress analysis, measured centre-of-gravity test, thermal test, battery runtime test, actuator torque validation, connector current validation, tolerance stack-up or physical prototype test is included in this report.

Engineering status: preliminary packaging concept — suitable for CAD review and iteration; not certified for fabrication or unsupervised operation.